QIANG LIU, MSc.

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RESEARCH INTERESTS

MRI image acquisition and reconstruction: open-source diffusion MRI and super-resolution sequence development.

EDUCATION

Ph.D. Candidate *Jul 2024 – Jun 2027 (expected)*

Northeastern University, Boston, MA

Advisor: Dr. Deniz Erdogmus (Northeastern University)

Co-Advisor: Dr. Yogesh Rathi (Harvard Medical School)

Visiting Student *May 2022 – May 2024*

Harvard Medical School, Boston, MA

Advisor: Dr. Yogesh Rathi (Harvard Medical School)

Research Topic: Open-source MRI sequence development and image reconstruction

M.S. in Biomedical Engineering *Sep 2020 – Jun 2024*

Southern Medical University, China

B.S. in Biomedical Engineering *Sep 2016 – Jun 2020*

Southern Medical University, China

AWARDS AND PRIZES

ISMRM ***Summa Cum Laude*** Merit Award 2026

Trainee (Educational) Stipend, ISMRM & ISMRT Annual Meeting 2023 – 2025

PUBLICATIONS

First-Author Publications

1. **Liu Q,** Wang Y, et al. High-SNR whole-brain mesoscale diffusion MRI using rotating-view acquisitions (ROVER-dMRI). ***In Revision.***
2. Tang L*, **Liu Q*,** et al. Scanner-adaptive coil-level denoising for diffusion MRI using explicit noise priors. ***2026 IEEE 23rd International Symposium on Biomedical Imaging (ISBI).*** 2026:1–5. (*Co-first authors)
3. **Liu Q,** Zhu A, et al. Effect of a consistent reconstruction algorithm on inter-scanner reproducibility in diffusion MRI. ***Medical Physics.*** 2025; 52:e70096.
4. **Liu Q,** Gagoski B, et al. Time-division multiplexing (TDM) sequence removes bias in T2 estimation and relaxation-diffusion measurements. ***Magnetic Resonance in Medicine.*** 2024; 92(6):2506–2519.
5. **Liu Q,** Ning L, et al. Reduced cross-scanner variability using vendor-agnostic sequences for single-shell diffusion MRI. ***Magnetic Resonance in Medicine.*** 2024; 92(1):246–256.
6. **Liu Q,** Xu Z, et al. Diffusion-weighted magnetic resonance imaging in rat kidney using two-dimensional navigated, interleaved echo-planar imaging at 7.0 T. ***NMR in Biomedicine.*** 2021; 34(4):e4652.

Co-authored Publications

1. Jeong H, **Liu Q,** Keenan KE, Wilde EA, Schneider W, Pathak S, Zuccolotto A, O'Donnell LJ, Ning L, Rathi Y. HARP: HARmonizing in-vivo single-shell diffusion MRI using phantom-only training. ***Magnetic Resonance in Medicine.*** 2026. doi:10.1002/mrm.70640
2. Yong X, Liu Z, Dong Y, Hong H, Chen Y, Dubovan P, **Liu Q,** Kim J, Dorken Gallastegi U, Jun Y, Bilgic B. SeqEyes: an open-source toolbox for synchronized waveform/trajectory visualization in Pulseq. ***Magnetic Resonance in Medicine.*** 2026.

3. Lyu J, Ning L, Consagra W, **Liu Q,** Rushmore RJ, Bilgic B, Rathi Y. Rapid whole brain motion-robust mesoscale in-vivo MR imaging using multi-scale implicit neural representation. ***Medical Image Analysis.*** 2026; 107:103830.
4. Jiang Y, Jun Y, **Liu Q,** Zhong W, et al. Multishot dual polarity GRAPPA: robust Nyquist ghost correction for multishot EPI. ***Magnetic Resonance in Medicine.*** 2026; 95(5):2726–2736.
5. Jun Y, **Liu Q,** Gong T, et al. PRIME: Phase reversed interleaved multi-Echo acquisition enables highly accelerated distortion-corrected diffusion MRI. ***Medical Image Analysis.*** 2026; 111:104058.
6. Zhang F, Chen Y, Ning L, Rushmore J, **Liu Q,** et al. Assessment of the depiction of superficial white matter using ultra-high-resolution diffusion MRI. ***Human Brain Mapping.*** 2024; 45(14):e70041.
7. Ruan G, Liu J, An Z, Wu K, Tong C, **Liu Q,** Liang P, et al. Automated skull stripping in mouse functional magnetic resonance imaging analysis using 3D U-Net. ***Frontiers in Neuroscience.*** 2022; 16:801769.
8. Yang Y, Li Z, **Liu Q,** Guo Y, et al. Carotid arterial wall MRI of apolipoprotein E-deficient mouse at 7 T using DANTE-prepared variable-flip-angle rapid acquisition with relaxation enhancement. ***Magnetic Resonance Imaging.*** 2022; 86:1–9.

SELECTED CONFERENCE ABSTRACTS

Liu Q, Wang Y, Pan Y, et al. High-SNR Whole-Brain Mesoscale Isotropic Diffusion MRI Using Rotating-View Acquisitions (ROVER-dMRI). ISMRM & SMRT Annual Meeting, 2026. **[Oral, AMPC Selected]**

Liu Q, Zhu A, Wang X, et al. Effect of Unified Reconstruction Algorithm on Inter-Scanner Variability in Diffusion MRI. ISMRM & SMRT Annual Meeting, 2025. **[Oral]**

Liu Q, Ning L, Shaik I, et al. Improved across-scanner reproducibility using vendor-agnostic diffusion sequences. ISMRM & SMRT Annual Meeting, 2024. **[Oral]**

Jun Y, **Liu Q,** Cho J, et al. PRIME: Phase Reversed Interleaved Multi-Echo acquisition enables highly accelerated distortion-free diffusion MRI. ISMRM & SMRT Annual Meeting, 2024. **[Oral]**

INVITED ORAL PRESENTATIONS

1. **Pulseq for High-Resolution Diffusion Acquisitions**, Vendor-Agnostic Pulse Sequence Programming with Pulseq II, ISMRM Virtual Meeting, February 2026.
2. **Vendor-Independent Harmonized gSlider for High-Isotropic-Resolution dMRI at Multiple Sites**, Vendor-Agnostic Pulse Sequence Programming & Image Reconstruction Member-Initiated Tutorials, ISMRM 2023, Toronto, Canada.

SELECTED RESEARCH PROJECTS

1. **High-SNR Whole-Brain Mesoscale Diffusion MRI Using Rotating-View Acquisitions (ROVER-dMRI)** ***(Under Review)***

   >>>> Developed ROVER-dMRI, a unified rotating-view acquisition and super-resolution reconstruction framework for high-SNR, whole-brain mesoscale diffusion MRI in clinically feasible scan times. Thick slices acquired across many rotated orientations boost baseline SNR, while a shared multi-resolution hash-encoded implicit neural representation with per-direction MLP heads and a physics-informed, Bloch-simulated slice-profile forward model reconstruct these views into a continuous isotropic volume (up to 13× super-resolution). Validated in vivo on human brain (0.78 and 0.5 mm isotropic) and monkey brain (0.6 mm) across scanners with differing gradient performance, and benchmarked against gSlider-SMS and least-squares reconstruction.

   >>>> The method preserved crossing-fiber structure, recovered fine cerebellar detail invisible at 0.78 mm, and cut scan time substantially while retaining fiber-orientation fidelity. Released open-source.

2. **Effect of a Consistent Reconstruction Algorithm on Inter-Scanner Reproducibility in Diffusion MRI** ***(Medical Physics, 2025)***

   >>>> Investigated how using a consistent image-reconstruction algorithm across scanners affects inter-scanner reproducibility in diffusion MRI, an underexplored factor in multi-center harmonization. Identical single-shell dMRI protocols were run on two clinical 3T systems from different vendors (Siemens Prisma and GE Premier), with five volunteers scanned repeatedly for within- and cross-scanner comparisons (20 scans total). Three reconstructions were assessed: vendor-provided online reconstruction, offline Split slice-GRAPPA, and offline L1-wavelet regularized SENSE. Microstructure (FA, MD from multi-tensor UKF tractography and DTI) and tractography measures (streamline counts, weighted Dice) were compared, with voxelwise noise estimation and bootstrapped uncertainty analysis.

   >>>> Offline Split-GRAPPA significantly reduced inter-scanner FA variability and achieved the lowest noise levels, though inter-scanner variability remained higher than within-scanner variability.

3. **Time-Division Multiplexing (TDM) Sequence Removes Bias in T2 Estimation and Relaxation-Diffusion Measurements** ***(Magnetic Resonance in Medicine, 2024)***

   >>>> Compared multi-echo (ME) and time-division multiplexing (TDM) sequences for accelerated relaxation-diffusion MRI, examining their reliability for accurate microstructure estimation. The ME, TDM, and reference single-echo (SE) sequences with six echo times were implemented in Pulseq under single-band and multi-band-2 acceleration. On a diffusion phantom, image intensities were compared via normalized RMS error, and Shinnar-Le Roux pulses were used to probe slice-profile effects on ME. In-vivo brain scans assessed sequence effects on T2 estimation and microstructure using REDIM and maximum-entropy relaxation-diffusion distribution methods.

   >>>> TDM performance closely matched the gold-standard SE acquisition, whereas ME showed substantially larger biases and underestimated T2. TDM delivered accurate relaxation-diffusion measurements while accelerating acquisition by a factor of two to three.

4. **Reduced Cross-Scanner Variability Using Vendor-Agnostic Sequences for Single-Shell Diffusion MRI** ***(Magnetic Resonance in Medicine, 2024)***

   >>>> Developed a vendor-neutral diffusion MRI pulse sequence on the open-source, vendor-agnostic Pulseq platform to reduce inter-scanner variability between different manufacturers. A standard EPI-based dMRI sequence was implemented in Pulseq and tested on two clinical scanners from different vendors (Siemens Prisma and GE Premier), systematically comparing within- and inter-scanner variability against vendor-provided sequences. Assessments covered a diffusion phantom and three human subjects, using standard error and Lin's concordance correlation to quantify repeatability and reproducibility of DTI metrics including fractional anisotropy and mean diffusivity.

   >>>> Running identical Pulseq sequences on both systems yielded more than a 2.5× reduction in phantom variability and markedly lower in-vivo variability, with over 50% reduction in cortical and subcortical regions, benefiting multi-center neuroimaging reproducibility.

5. **Diffusion-Weighted MRI in Rat Kidney Using 2D Navigated, Interleaved Echo-Planar Imaging at 7.0 T** ***(NMR in Biomedicine, 2021)***

   >>>> Investigated the feasibility of two-dimensional navigated, interleaved multishot echo-planar imaging (EPI) to improve kidney diffusion-weighted imaging in rats at 7.0 T. A fully sampled interleaved four-shot EPI sequence with 2D navigators was tailored for rat kidney DWI on a preclinical small-bore scanner. Image quality of four-shot EPI was compared against T2-weighted RARE (reference) and single-shot EPI with and without parallel imaging, using contrast-to-noise ratio, Dice similarity, and Hausdorff distance to evaluate distortion. Mean diffusivity and fractional anisotropy were computed for renal cortex and medulla.

   >>>> Four-shot EPI achieved the highest CNR and lowest geometric distortion among the EPI variants, clearly identified the anisotropic inner stripe of the outer medulla, and discerned corticomedullary differences in both MD and FA.

6. **PRIME: Phase-Reversed Interleaved Multi-Echo Acquisition for Highly Accelerated Distortion-Corrected Diffusion MRI** ***(Second Author; Medical Image Analysis, 2026)***

   >>>> Contributed to PRIME, a phase-reversed interleaved multi-echo acquisition enabling highly accelerated, distortion-corrected diffusion MRI without prolonging TR when generalized slice-dithered enhanced resolution (gSlider) encoding is used for volumetric acquisition. The sequence acquires a first echo targeting diffusion-weighted imaging plus additional echoes at lower or higher resolution for high-fidelity field-map estimation, shot-to-shot phase navigation, motion navigation, or diffusion relaxometry.

   >>>> Evaluated in vivo on clinical and Connectome 2.0 scanners, PRIME demonstrated high in-plane acceleration using field maps from a second echo, high-resolution diffusion relaxometry parameters estimated via a white-matter multi-TE spherical-mean model, and high-fidelity mesoscale diffusion imaging at 490 micron isotropic resolution by leveraging high-performance gradients, all without added scan time under gSlider encoding.

TECHNICAL SKILLS

Programming: Python, C/C++, MATLAB

Deep Learning: PyTorch (implicit neural representations, multi-resolution hash encoding, physics-informed forward models, MR denoising); GPU-accelerated training and optimization

MRI Reconstruction: Parallel imaging (GRAPPA, Split slice-GRAPPA, SENSE), regularized reconstruction (L1-wavelet),

super-resolution reconstruction, EPI ghost and distortion correction

MRI Sequence Programming: Pulseq (vendor-agnostic; Siemens and GE scanners), Philips, Bruker

Software Engineering: Git/GitHub (open-source code releases), Linux